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PAPER SOLUTION
FOR
DDCET - 2025**

SUBJECTS:

Basics of Science and Engineering

&

Aptitude Test (Mathematics and Soft Skill)

Prepared by:

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Subject wise analysis

Subject	Number of questions
Basics of Science and Engineering (DDCET)	50
Aptitude Test (Mathematics and Soft Skill) DDCET	50
Total	100

Q1. A big piece of glass is first heated and then is allowed to cool. On cooling down, a crack is developed in it. One of the possible reasons for this is _____.

- A. High melting point
- B. Large thermal conductivity
- C. Large specific heat
- D. Small thermal conductivity

Correct Ans: D

Solution :

⇒ When glass cools unevenly due to **small thermal conductivity**, thermal stress develops, causing cracks. Mathematically, thermal stress $\sigma = E\alpha\Delta T$, where E is Young's modulus, α is thermal expansion coefficient, and ΔT is temperature gradient.

Q2. When heat is given to boiling water, the temperature of water _____.

- A. increases
- B. decreases
- C. remains constant
- D. None of the above.

Correct Ans: C

Solution :

⇒ **Boiling** water maintains constant temperature (100°C at 1 atm) because heat energy ($Q = mL$) is used for phase change (latent heat of vaporization L), not increasing kinetic energy. Thus, $\Delta T = 0$.

Q3. $37^{\circ}\text{C} = \text{_____}^{\circ}\text{F}$.

- A. 98.6
- B. 98.8
- C. 97.6
- D. 96.8

Correct Ans: A

Solution :

⇒ Convert $37^{\circ}C$ to Fahrenheit: $T_F = \frac{9}{5}T_C + 32 = \frac{9}{5} \times 37 + 32 = 98.6^{\circ}F$.

Q4. The SI Unit of coefficient of thermal conductivity is _____ .

- A. J/m K
- B. N/m K
- C. W/mK
- D. mK/W

Correct Ans: C

Solution :

⇒ Thermal conductivity (k) measures heat transfer rate per unit area and temperature gradient. SI unit: W/mK (Watt per meter per Kelvin). Derived from Fourier's law: $Q = -kA \frac{dT}{dx}$.

Q5. Audible waves have a frequency range from _____ .

- A. 20 Hz-10 KHz
- B. 20 Hz-20 KHz
- C. 10 Hz-10 KHz
- D. 10 Hz - 20 KHz

Correct Ans: B

Solution :

⇒ Audible frequency range for humans: 20 Hz (low-pitched) to 20 kHz (high-pitched). Ultrasonic waves (> 20 kHz) and infrasonic waves (< 20 Hz) are inaudible.

Q6. The speed of light in a vacuum is _____ .

- A. 3×10^5 m/s
- B. 3×10^6 m/s
- C. 3×10^7 m/s
- D. 3×10^8 m/s

Correct Ans: D

Solution :

⇒ Speed of light in vacuum (c) is a fundamental constant:
 $c = 299,792,458 \text{ m/s} \approx 3 \times 10^8 \text{ m/s}$.

Q7. What is the periodic time of a wave having frequency of 20 Hz?

- A. 0.05 sec
- B. 0.5 sec
- C. 5 sec
- D. 50 sec

Correct Ans: A

Solution :

⇒ Period (T) and frequency (f) are inversely related: $T = \frac{1}{f} = \frac{1}{20 \text{ Hz}} = 0.05 \text{ sec.}$

Q8. LASER is an extremely _____ radiation.

- A. Unidirectional
- B. Focused
- C. Coherent and Stimulated
- D. All of above

Correct Ans: D

Solution :

⇒ LASER properties: **Unidirectional** - travels in one direction, **Focused** - narrow beam, **Coherent** in-phase waves, and **Stimulated** - emission. Hence, all options are correct.

Q9. A volume of an auditorium is 5000 m^3 . The total absorption by the surfaces is 900 O.W.U. Then calculate reverberation time.

- A. 8.094
- B. 0.729
- C. 0.894
- D. 1.003

Correct Ans: C

Solution :

⇒ Reverberation time (T) is calculated using Sabine's formula: $T = \frac{0.161V}{A}$, where $V = 5000 \text{ m}^3$ (volume) and $A = 900 \text{ OWU}$ (absorption). Substituting:
 $T = \frac{0.161 \times 5000}{900} = 0.894 \text{ sec.}$

Q10. If n_1 and n_2 are refractive indices of core and cladding resp. Then what is the formula for numerical aperture for optical fiber?

- A. $\sqrt{n_2^2 - n_1^2}$
- B. $\sqrt{n_1^2 - n_2^2}$

- C. $\sqrt{n_2 - n_1}$
 D. $\sqrt{n_1 - n_2}$

Correct Ans: B

Solution :

$\Rightarrow$ Numerical Aperture (NA) for optical fiber: $NA = \sqrt{n_1^2 - n_2^2}$, where n_1 (core) and n_2 (cladding) are refractive indices. This determines light-gathering capacity.

Q11. $CaCO_{3(s)} \xrightarrow{\text{heat}} CaO_{(s)} + CO_{2(g)}$, which kind of reaction is this?

- A. Displacement reaction
 B. Oxidation reaction
 C. Thermal decomposition reaction
 D. Precipitation reaction

Correct Ans: C

Solution :

$\Rightarrow$ Here, a single element $CaCO_3$ breaks down into two product that is CaO and CO_2 .

$\Rightarrow$ Hence, this is a decomposition process.

Q12. What is the formula of lead sulphate ?

- A. Pb_2SO_4
 B. $Pb(SO_4)_2$
 C. $PbSO_4$
 D. $Pb_2(SO_4)_3$

Correct Ans: C

Solution :

$\Rightarrow$ The formula of lead sulphate is: $PbSO_4$

where,

$Pb = \text{Lead}$

$SO_4 = \text{Sulphate ion}$

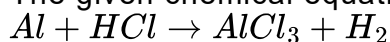
Q13. Which is the balance equation of $Al + HCl \rightarrow AlCl_3 + H_2$?

- A. $Al + HCl \rightarrow AlCl_3 + 2H_2$
 B. $2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2$
 C. $2Al + 3HCl \rightarrow 2AlCl_3 + 2H_2$
 D. $2Al + 4HCl \rightarrow 3AlCl_3 + 3H_2$

Correct Ans: B

Solution :

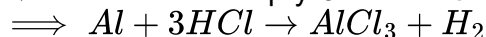
⇒ The given chemical equation is



⇒ Here, the number of atoms of hydrogen and chlorine is not equal on both the side.

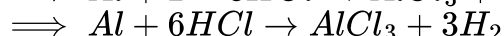
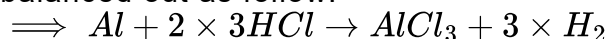
Step 1: Balance *Cl* both side as it has maximum atoms.

⇒ To do this multiply 3 with *HCl* on the reactant side



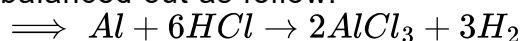
Step 2: Balance *H* both side.

⇒ Reactant have 3 atoms and product have 2 atoms of *H*, which can be balanced out as follow:



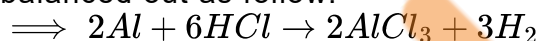
Step 3: Now, we need to balance *Cl* again.

⇒ Reactant have 6 atoms and product have 3 atoms of *Cl*, which can be balanced out as follow:



Step 4: Finally, we need to balance *Al*.

⇒ Reactant have 1 atoms and product have 2 atoms of *Al*, which can be balanced out as follow:



Q14. Which main product is obtained by reaction of sodium hydroxide with zinc metal?

- A. $NaZnO_2$
- B. $NaZn_2O$
- C. Na_2ZnO_2
- D. $Zn(OH)_2$

Correct Ans: C

Solution :

⇒ Sodium zincate - Na_2ZnO_2 .

⇒ It is formed when zinc reacts with sodium hydroxide ($NaOH$) in an aqueous solution.

⇒ It is often used in galvanizing and electroplating processes and can also be found in some types of batteries.

Q15. Which type of oxides is called basic oxide ?

- A. Non-metallic oxide
- B. Metallic oxide
- C. (A) and (B) both
- D. Inactive oxide

Correct Ans: B

Solution :

⇒ Basic oxides are metal oxides that react with water to form a base or with acids to form a salt and water.

⇒ They are typically ionic compounds formed by the reaction of metals with oxygen.

Q16. Which of the following is known as milk of magnesia?

- A. $Mn(OH)_2$
- B. $Mg(OH)_2$
- C. MgO
- D. $Mg(OH)$

Correct Ans: B

Solution :

⇒ Magnesium hydroxide ($Mg(OH)_2$) - a milky white liquid is known as milk of magnesia.

Q17. Which of the following is the chemical formula of bleaching powder?

- A. $Ca(OH)_2$
- B. $CaOCl_2$
- C. Ca_2OCl
- D. $CaCO_3 \cdot 2H_2O$

Correct Ans: B

Solution :

⇒ Bleaching powder is also known as calcium hypochlorite, and its formula is $CaOCl_2$.

Q18. Which of the following is the lustrous non-metal?

- A. Iodine
- B. Carbon
- C. Nitrogen
- D. Bromine

Correct Ans: A

Solution :

⇒ Iodine is non metal having lustrous appearance.

Q19. Which of the following metals neither reacts with cold nor hot water, but reacts with vapour?

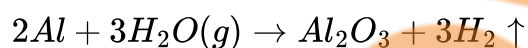
- A. Aluminium
- B. Iron
- C. Zinc
- D. All of the above

Correct Ans: D

Solution :

The correct answer is: Aluminium (Al)

⇒ Among metals, aluminium does not react with cold or hot water due to the formation of a passive oxide layer Al_2O_3 on its surface. However, it does react with steam (water vapour) at high temperatures to form aluminium oxide and hydrogen gas:



Other metals comparison:

- Sodium (Na) and Potassium (K): React vigorously with cold water.
- Calcium (Ca) and Magnesium (Mg): React with cold or hot water.
- Iron (Fe): Reacts slowly with steam.
- Zinc (Zn): Also reacts with steam.

∴ Aluminium fits the condition best — no reaction with cold or hot water, but reacts with steam

Q20. Which of the following is correct order of reactivity of metals?

- A. $Mg > Al > Zn > Fe$
- B. $Mg > Fe > Zn > Al$
- C. $Fe > Zn > Al > Mg$
- D. $Mg > Al > Fe > Zn$

Correct Ans: A

Solution :

K	Potassium		Most reactive Reactivity Least reactive
Na	Sodium		
Ca	Calcium		
Mg	Magnesium		
Al	Aluminium		
Zn	Zinc		
Fe	Iron		
Sn	Tin		
Pb	Lead		
Cu	Copper		
Hg	Mercury		
Ag	Silver		
Au	Gold		

⇒ The figure shows the list of elements arranged according to their reactivity.

Q21. The smallest unit of data in a computer is called a:

- A. Byte
- B. Bit
- C. Nibble
- D. GB

Correct Ans: B

Solution :

⇒ A single bit can only have two possible values: 0 or 1. Computers rely on these values to store and handle information.

⇒ Bits are usually combined to create bigger units of data like bytes, which consist of 8 bits, or kilobytes, which contain 1024 bytes.

Q22. Pressing Ctrl + Home keys on the Keyboard moves the cursor to the _____ in MS-Word.

- A. End of the document
- B. Beginning of the line
- C. Beginning of the paragraph

D. Beginning of the document

Correct Ans: D

Solution :

In MS Word, Pressing

- ⇒ Ctrl + End moves the cursor to the end of the document .
- ⇒ Only Home moves it to the beginning of the current line.
- ⇒ Ctrl + Up arrow(↑) jumps to the beginning of the current paragraph.
- ⇒ Ctrl + Home returns to the very start of the document.

Q23. Which of the following can be classified as a relative link in the HTML?

- A.
- B.
- C.
- D. All of the above

Correct Ans: B

Solution :

- ⇒ Relative URLs omit the scheme/host and specify a path relative to the current document (or its parent directories).
- ⇒ Absolute URLs include a scheme (e.g. http:// or https://) and usually a host name.

Q24. To find the total value of the cell values in A2, B2, C2 and D2 in MS-Excel, the correct formula is:

- A. sum (A2:D2)
- B. sum (A2, B2, C2, D2)
- C. Both (A) and (B)
- D. None of the above

Correct Ans: C

Solution :

The correct choice is Both (A) and (B).

- ⇒ =SUM(A2:D2) uses a continuous range from A2 through D2.
- ⇒ =SUM(A2, B2, C2, D2) explicitly lists each cell and add them up.

Q25. Which of the following is an incorrect option in the view menu of a Microsoft Power point presentation ?

- A. Tree view
- B. Outline view
- C. Notes page
- D. Reading view

Correct Ans: A

Solution :

In MS Powerpoint,

⇒ Outline view shows only the text content of your slides in a collapsible, linear outline on the left pane.

⇒ Notes page displays a single slide at the top and its speaker notes below, formatted as a printable "notes page."

⇒ Reading view runs your presentation in a window (not full-screen) so you can step through slides as a viewer would.

⇒ PowerPoint doesn't include any option in view menu named Tree View.

Q26. The green plants manufacture their own food are known as _____.

- A. Heterotrophs
- B. Autotrophs
- C. Saprotrophs
- D. Omnivores

Correct Ans: B

Solution :

⇒ Autotrophs are organisms capable of producing their own food, often through photosynthesis, using light energy, water, and carbon dioxide.

Q27. Bag filters are used to _____.

- A. reduce noise pollution
- B. reduce sludge from waste water
- C. remove particulate matter for a contaminated gas stream
- D. reduce soil pollution

Correct Ans: C

Solution :

⇒ A bag filter is an air pollution control device that uses fabric bags to trap particulate matter (dust and other small particles) from polluted gas streams.

⇒ It's commonly used in industrial settings like boiler houses, factories, and power plants to reduce emissions and improve air quality.

Q28. The process of producing energy by utilizing heat trapped inside the earth surface is known as _____.

- A. Geo-Thermal energy
- B. Hydrothermal energy
- C. Solar energy
- D. Wave energy

Correct Ans: A

Solution :

⇒ The process of producing energy by utilizing heat trapped inside the Earth's surface is called geothermal energy.

Q29. _____ is a device that converts solar energy directly into electricity.

- A. Fuel cell
- B. Hydrogen cell
- C. Solar array
- D. Photovoltaic cell

Correct Ans: D

Solution :

⇒ A solar cell (also known as a photovoltaic cell) is a device that converts solar energy directly into electricity.

Q30. The goal of the Paris Agreement is to keep the increase in global temperature well below _____ above Preindustrial levels.

- A. 1.0°C
- B. 1.5°C
- C. 2.0°C
- D. 2.5°C

Correct Ans: C

Solution :

⇒ The Paris Agreement aims to keep the increase in global temperature well below 2 degrees Celsius.

Q31. Number of significant digits in 2025 is _____ .

- A. 3
- B. 4
- C. 2
- D. 5

Correct Ans: B

Solution :

⇒ Significant digits in 2025: All non-zero digits (2, 0, 2, 5) are significant. Zero (0) between non-zero digits is also significant.

Q32. Zero error is an example of _____ error.

- A. Systematic
- B. Random
- C. Maximum
- D. Minimum

Correct Ans: A

Solution :

⇒ Zero error is **systematic** because it arises from instrument calibration (e.g., micrometer screw gauge not starting at zero) and is repeatable. Random errors vary unpredictably.

Q33. 1 joule = _____ erg.

- A. 10^3
- B. 10^5
- C. 10^7
- D. 10^9

Correct Ans: C

Solution :

⇒ Energy unit conversion: $1 \text{ J} = 10^7 \text{ erg}$.

Derived from $1 \text{ J} = 1 \text{ kg m}^2/\text{s}^2$ and $1 \text{ erg} = 1 \text{ g cm}^2/\text{s}^2$.

Q34. Density of water in CGS system is _____.

- A. 1000
- B. 100
- C. 10
- D. 1

Correct Ans: D

Solution :

⇒ Density of water in CGS: $\rho = 1 \text{ g/cm}^3$ (at 4°C). In SI: 1000 kg/m^3 .

Q35. In equation density $s = \frac{m}{V}$ if percentage error in mass m is 0.3% and for radius r is 0.5% then percentage error in density of a Sphere is _____%.

- A. 1.8
- B. 1.5
- C. 1.2
- D. 1.1

Correct Ans: A

Solution :

⇒ For a sphere ($V = \frac{4}{3}\pi r^3$), 1% error in density (s):
 $\Delta s\% = \Delta m\% + 3 \times \Delta r\% = 0.3\% + 3 \times 0.5\% = 1.8\%$.

Q36. The pitch of micrometer screw is 1 mm. There are 100 divisions on it's circular scale. What will be the least count of the micrometer screw.

- A. 1 mm
- B. 0.1 mm
- C. 0.01 mm
- D. 0.001 mm

Correct Ans: C

Solution :

⇒ Least Count = $\frac{\text{Pitch}}{\text{Divisions}} = \frac{1 \text{ mm}}{100} = 0.01 \text{ mm}$. Ensures precision in micrometer measurements.

Q37. The SI Unit of linear momentum is _____ .

- A. Kgs/m
- B. Kg m/s
- C. ms/Kg
- D. Kg s m

Correct Ans: B

Solution :

⇒ Linear momentum: $p = mv$. SI unit: kg m/s.

Q38. The centripetal force always acts _____ .

- A. Away from the centre
- B. In tangential direction
- C. Perpendicular to the motion
- D. Towards the centre

Correct Ans: D

Solution :

⇒ Centripetal force ($F_c = \frac{mv^2}{r}$) always points **towards the centre** of circular motion, perpendicular to velocity.

Q39. When external force acting on a body is zero then its acceleration is _____ .

- A. increases
- B. decreases
- C. equal to zero
- D. remains constant

Correct Ans: C

Solution :

⇒ Newton's First Law: If net force $F = 0$, acceleration $a = 0$ (object remains at rest or moves uniformly).

Q40. The formula for angular velocity of a body moving with a constant speed v in a circle of radius r , is given by _____

- A. v/r
- B. vr
- C. vr^2
- D. vr^3

Correct Ans: A

Solution :

⇒ Angular velocity (ω) for circular motion: $\omega = \frac{v}{r}$ (radians/second).

Q41. A body having 10 kg mass and 9.8 m/s^2 acceleration produces how much force ?

- A. 9.8 newton
- B. 98 newton
- C. 0.98 newton
- D. 980 newton

Correct Ans: B

Solution :

⇒ Force (F) via Newton's Second Law:

$$F = ma = 10 \text{ kg} \times 9.8 \text{ m/s}^2 = 98 \text{ N}.$$

Q42. If an object having mass 50 kg rotated in a circle with a radius of 2 m with a constant speed of 60 m/s then what is the value of centripetal acceleration?

- A. 6000 m/s^2
- B. 3000 m/s^2
- C. 1800 m/s^2
- D. 1500 m/s^2

Correct Ans: C

Solution :

$\Rightarrow$ Centripetal acceleration (a_c): $a_c = \frac{v^2}{r} = \frac{(60 \text{ m/s})^2}{2 \text{ m}} = 1800 \text{ m/s}^2$.

Q43. The SI Unit of electric flux is _____ .

- A. C^2/Nm
- B. Nm/C
- C. Nm/C^2
- D. Nm^2/C

Correct Ans: D

Solution :

$\Rightarrow$ Electric flux (Φ) SI unit: Nm^2/C .
From Gauss's law: $\Phi = \oint \mathbf{E} \cdot d\mathbf{A}$.

Q44. The electric field lines are _____ .

- A. Imaginary
- B. Real
- C. Ideal
- D. None of the above

Correct Ans: A

Solution :

$\Rightarrow$ Electric field lines are **imaginary** curves representing the direction of the electric field (**E**) at each point.

Q45. How many electrons are in 1 coulomb charge ?

- A. 6.25×10^{17}
- B. 6.25×10^{18}
- C. 6.25×10^{19}
- D. 6.25×10^{20}

Correct Ans: B

Solution :

$\Rightarrow$ Number of electrons in 1 Coulomb: $n = \frac{Q}{e} = \frac{1 \text{ C}}{1.6 \times 10^{-19} \text{ C}} = 6.25 \times 10^{18}$.

Q46. The capacitance of a capacitor _____ if an insulating dielectric material is placed between two plates of it.

- A. increases
- B. decreases
- C. remains constant
- D. becomes zero

Correct Ans: A

Solution :

⇒ Capacitance (C) with dielectric: $C = \kappa C_0$, where κ (dielectric constant) > 1 .
Thus, C **increases**.

Q47. If the distance between two charges is doubled then electric force between them becomes _____.

- A. half
- B. double
- C. one fourth
- D. four times

Correct Ans: C

Solution :

⇒ Coulomb's Law: $F \propto \frac{1}{r^2}$. If distance r doubles, force reduces to $\frac{1}{4}$.

Q48. What is the effective value of resistance when three resistors $5\ \Omega$, $10\ \Omega$ and $15\ \Omega$ are connected in series combination?

- A. $10\ \Omega$
- B. $30\ \Omega$
- C. $20\ \Omega$
- D. $5.5\ \Omega$

Correct Ans: B

Solution :

⇒ Series resistance: $R_{\text{total}} = R_1 + R_2 + R_3 = 5\ \Omega + 10\ \Omega + 15\ \Omega = 30\ \Omega$.

Q49. Heat capacity of a body per unit mass is known as _____.

- A. Specific heat
- B. Relative heat
- C. C_p
- D. C_v

Correct Ans: A

Solution :

⇒ Specific heat capacity (c): Heat required per unit mass to raise temperature by 1°C . SI unit: J/kg K .

Q50. 1 KCal= _____ J.

- A. 4.186
- B. 4186
- C. 41.86
- D. 418.6

Correct Ans: B

Solution :

⇒ Energy conversion: $1 \text{ kcal} = 4186 \text{ J}$ (exact: $1 \text{ cal} = 4.186 \text{ J}$).

Q51. The Centre of the circle $(x + 3)^2 + y^2 - 2 = 0$ is_____.

- A. (3,0)
- B. (0,3)
- C. (-3,0)
- D. (0,-3)

Correct Ans: C

Solution :

$$(x + 3)^2 + y^2 - 2 = 0$$

$$(x + 3)^2 + y^2 = 2$$

Standard form:

$$(x - h)^2 + (y - k)^2 = r^2$$

Where:

- (h, k) is the centre of the circle
- r is the radius

Compare:

$$(x + 3)^2 + y^2 = (x - (-3))^2 + (y - 0)^2$$

$$\therefore h = -3 \text{ \& } k = 0$$

$\therefore$ The centre of the circle is $(-3, 0)$.

Q52. If $f(x) = e^{\log x}$, then $f(1) = \underline{\hspace{2cm}}$

- A. 0
- B. 1
- C. e
- D. -e

Correct Ans: B

Solution :

$$f(x) = e^{\log x}$$

where, $e^{\log_e x} = x$ (for $x > 0$ and natural log)

$$\therefore f(x) = x$$

$$\therefore f(1) = 1$$

Q53. $\lim_{x \rightarrow 0} \frac{\sin 2x}{\tan x} = \underline{\hspace{2cm}}$

- A. 1
- B. 0.5
- C. 2
- D. 3

Correct Ans: C

Solution :

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{\tan x}$$

$$\sin 2x = 2 \sin x \cos x \quad \text{and} \quad \tan x = \frac{\sin x}{\cos x}$$

$$\frac{2 \sin x \cos x}{\frac{\sin x}{\cos x}} = 2 \sin x \cos x \cdot \frac{\cos x}{\sin x}$$

$$= 2 \cancel{\sin x} \cos x \cdot \frac{\cos x}{\cancel{\sin x}} = 2 \cos^2 x$$

Take the limit as $x \rightarrow 0$

$$2 \cos^2 x = 2 \cdot 1 = 2$$

Q54. $\lim_{n \rightarrow \infty} \left(1 + \frac{5}{2n}\right)^{8n} = \underline{\hspace{2cm}}$

- A. e^{13}
- B. e^4
- C. e^{10}
- D. e^{20}

Correct Ans: D

Solution :

$$\lim_{n \rightarrow \infty} \left(1 + \frac{5}{2n}\right)^{8n}$$

→ Standard exponential limit identity:

$$\lim_{n \rightarrow \infty} \left(1 + \frac{k}{n}\right)^n = e^k$$

→ Rewrite the expression:

$$\left(1 + \frac{5}{2n}\right)^{8n} = \left[\left(1 + \frac{5}{2n}\right)^{2n}\right]^4$$

$$\left(1 + \frac{5}{2n}\right)^{2n} \rightarrow e^5 \quad \text{as } n \rightarrow \infty$$

$$\therefore \left[\left(1 + \frac{5}{2n}\right)^{2n}\right]^4 \rightarrow (e^5)^4 = e^{20}$$

Q55. $\frac{d}{dx}(\log 5 + \sin \frac{\pi}{3}) = \underline{\hspace{2cm}}$

- A. 0
- B. $\frac{1}{5} + \cos \frac{\pi}{3}$
- C. $\frac{1}{5} - \cos \frac{\pi}{3}$
- D. $\cos \frac{\pi}{3}$

Correct Ans: A

Solution :

$$\frac{d}{dx} \left(\log 5 + \sin \frac{\pi}{3} \right)$$

→ $\log 5$ is a constant (natural log of 5)

→ $\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$, also a constant

∴ Derivative of a constant is 0

$$\frac{d}{dx}(\text{constant}) = 0$$

Q56. $\frac{d}{dx}(x^e + e^x - e^e) = \underline{\hspace{2cm}}$

- A. e^x
- B. $ex^e + e^x$
- C. $ex^{e-1} + e^x$
- D. $ex^{e-1} + e^x - e^{e-1}$

Correct Ans: C

Solution :

$$\frac{d}{dx}(x^e + e^x - e^e)$$

Differentiate each term:

1. $\frac{d}{dx}(x^e) = ex^{e-1}$

(Since e is a constant exponent)

$$2. \frac{d}{dx}(e^x) = e^x$$

$$3. \frac{d}{dx}(e^e) = 0$$

(Because it is a constant)

$$\frac{d}{dx}(x^e + e^x - e^e) = ex^{e-1} + e^x$$

Q57. $\frac{d}{dx}\log(\sin x) = \underline{\hspace{2cm}}$

- A. $\cot x$
- B. $-\cot x$
- C. $\tan x$
- D. $-\tan x$

Correct Ans: A

Solution :

$$\frac{d}{dx}\log(\sin x)$$

→ Apply the chain rule:

The derivative of $\log(u)$ with respect to x is:

$$\frac{d}{dx}\log(u) = \frac{1}{u} \cdot \frac{du}{dx}$$

Here, $u = \sin x$.

$$\frac{d}{dx}\sin x = \cos x$$

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$$\frac{d}{dx} \log(\sin x) = \frac{1}{\sin x} \cdot \cos x$$

$$\frac{\cos x}{\sin x} = \cot x$$

Q58. If $x + y = -2$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$

- A. 0
- B. 1
- C. 2
- D. -1

Correct Ans: D

Solution :

$$x + y = -2$$

$\Rightarrow$ Differentiate both sides:

$$\rightarrow \frac{d}{dx}(x + y) = \frac{d}{dx}(x) + \frac{d}{dx}(y) = 1 + \frac{dy}{dx}$$

$$\rightarrow \frac{d}{dx}(-2) = 0$$

$$\therefore 1 + \frac{dy}{dx} = 0$$

Q59. $\int \frac{1}{e^x} dx = \underline{\hspace{2cm}} + C$

- A. e^{-x}
- B. $-e^{-x}$
- C. e^x
- D. $-e^x$

Correct Ans: B

Solution :

$$\int \frac{1}{e^x} dx$$

$$\text{where, } \frac{1}{e^x} = e^{-x}$$

$$\int e^{-x} dx$$

→ The integral of e^{ax} with respect to x is $\frac{e^{ax}}{a}$, where a is a constant. Here, $a = -1$.

$$\therefore \int e^{-x} dx = \frac{e^{-x}}{-1} = -e^{-x}$$

$$\therefore \int \frac{1}{e^x} dx = -e^{-x} + C$$

Q60. $\int \frac{3}{5x^4} dx = \text{---} + C$

- A. $\frac{-3}{5x^3}$
- B. $\frac{-12}{5x^5}$
- C. $\frac{3}{20x^3}$
- D. $\frac{-1}{5x^3}$

Correct Ans: D

Solution :

$$\int \frac{3}{5x^4} dx$$

$$\int \frac{3}{5} x^{-4} dx = \frac{3}{5} \int x^{-4} dx$$

Use the power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} \quad \text{for } n \neq -1$$

Here, $n = -4$, so:

$$\int x^{-4} dx = \frac{x^{-3}}{-3} = -\frac{1}{3x^3}$$

$$\therefore \frac{3}{5} \cdot \left(-\frac{1}{3x^3} \right) = -\frac{1}{5x^3}$$

Q61. $\int \frac{4x-10}{x^2-5x+1} dx = \underline{\hspace{2cm}} + C$

- A. $2\log|2x - 5|$
- B. $2\log|x^2 - 5x + 1|$
- C. $\frac{2}{\log|2x-1|}$
- D. $\frac{2}{\log|x^2-5x+1|}$

Correct Ans: B

Solution :

$$\int \frac{4x - 10}{x^2 - 5x + 1} dx$$

Let:

$$u = x^2 - 5x + 1 \Rightarrow du = (2x - 5) dx$$

$$\therefore \frac{4x - 10}{x^2 - 5x + 1} dx = \frac{2(2x - 5)}{u} dx = 2 \cdot \frac{du}{u}$$

$$\int 2 \cdot \frac{1}{u} du = 2 \ln |u| + C$$

Substitute back: $u = x^2 - 5x + 1$:

$$= 2 \ln |x^2 - 5x + 1| + C$$

Q62. $\int_{-\pi}^{\pi} x^2 \sin x \, dx = \underline{\hspace{2cm}}$

- A. 0
- B. 1
- C. -1
- D. -2π

Correct Ans: A

Solution :

$$\int_{-\pi}^{\pi} x^2 \sin x \, dx$$

$$\rightarrow x^2 \text{ is even: } x^2 = (-x)^2$$

$$\rightarrow \sin x \text{ is odd: } \sin(-x) = -\sin x$$

$\therefore x^2 \sin x$ is the product of even and odd functions $\Rightarrow$ an odd function

$\rightarrow$ Use symmetry of definite integrals.

$\rightarrow$ The integral of an odd function over a symmetric interval $[-a, a]$ is always zero:

$$\int_{-a}^a (\text{odd function}) dx = 0$$

Q63. $\frac{\log 125}{\log 5} = \underline{\hspace{2cm}}$

- A. $\log 125 - \log 5$
- B. $\log 120$
- C. 25
- D. 3

Correct Ans: D

Solution :

$$\frac{\log 125}{\log 5}$$

where, $\log 125 = \log(5^3) = 3 \log 5$

$$\frac{\log 125}{\log 5} = \frac{3 \log 5}{\log 5}$$

$$= 3$$

Q64. $\log 2 + 2 \log 3 = \underline{\hspace{2cm}}$

- A. $2\log 6$
- B. $\log 8$
- C. $\log 18$
- D. $\log 12$

Correct Ans: C

Solution :

$$\log 2 + 2 \log 3$$

$$\text{where, } 2 \log 3 = \log(3^2) = \log 9$$

$$\therefore \log 2 + \log 9 = \log(2 \cdot 9) = \log 18$$

Q65. The mean of the first five negative integers is ____.

- A. 15
- B. -15
- C. 3
- D. -3

Correct Ans: D

Solution :

The first five negative integers are:

$$-1, -2, -3, -4, -5$$

$$\therefore \text{Mean} = \frac{\text{sum}}{\text{number of terms}} = \frac{-15}{5} = -3$$

Q66. $\begin{bmatrix} \tan x & -\csc x \\ -\sin x & -\cot x \end{bmatrix} = \underline{\hspace{2cm}}$

- A. 0
- B. -1
- C. -2
- D. -3

Correct Ans: C

Solution :

$$\begin{bmatrix} \tan x & -\csc x \\ -\sin x & -\cot x \end{bmatrix}$$

$$\therefore \det = (\tan x)(-\cot x) - (-\csc x)(-\sin x)$$

$$\therefore \det = \left(\frac{\sin x}{\cos x} \right) \left(-\frac{\cos x}{\sin x} \right) - \left(-\frac{1}{\sin x} \right) (-\sin x)$$

$$\therefore \det = -1 - 1 = -2$$

Q67.

The matrix $\begin{bmatrix} 3 & 1 & -2 \\ 1 & 0 & 5 \\ -2 & 5 & 4 \end{bmatrix}$ is a ____ matrix.

- A. Singular
- B. Symmetric
- C. Skew-symmetric
- D. Diagonal

Correct Ans: B

Solution :

$$\text{Given : } A = \begin{bmatrix} 3 & 1 & -2 \\ 1 & 0 & 5 \\ -2 & 5 & 4 \end{bmatrix}$$

1. Symmetric Matrix:

A matrix is symmetric if $A = A^T$ (transpose equals the original).

$$A^T = \begin{bmatrix} 3 & 1 & -2 \\ 1 & 0 & 5 \\ -2 & 5 & 4 \end{bmatrix} = A$$

∴ A is symmetric.

2. Skew-symmetric Matrix:

For a matrix to be **skew-symmetric**, $A^T = -A$.

Here, since $A = A^T$, it is **not** skew-symmetric.

3. Diagonal Matrix:

A diagonal matrix has non-zero entries only on the main diagonal and zeros elsewhere.

This matrix has non-zero off-diagonal elements (like 1 and -2), so it's not diagonal.

4. Singular Matrix:

A matrix is singular if its determinant is 0.

$$\det(A) = 3 \begin{vmatrix} 0 & 5 \\ 5 & 4 \end{vmatrix} - 1 \begin{vmatrix} 1 & 5 \\ -2 & 4 \end{vmatrix} + (-2) \begin{vmatrix} 1 & 0 \\ -2 & 5 \end{vmatrix}$$

$$\det(A) = -75 - 14 - 10 = -99 \neq 0$$

∴ It is not singular.

Q68.

If $A = \begin{bmatrix} 3 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -5 \\ 0 \\ 4 \end{bmatrix}$ then $AB = \underline{\hspace{2cm}}$

- A. $\begin{bmatrix} -2 & 1 & 6 \end{bmatrix}$
- B. $\begin{bmatrix} -15 & 0 & 8 \end{bmatrix}$
- C. $\begin{bmatrix} -7 \end{bmatrix}$
- D. *None*

Correct Ans: C

Solution :

$$A = [3 \quad 1 \quad 2], \quad B = \begin{bmatrix} -5 \\ 0 \\ 4 \end{bmatrix}$$

→ A is a 1×3 row matrix, and B is a 3×1 column matrix,

∴ The product AB is defined and will result in a 1×1 matrix.

$$\therefore AB = (3)(-5) + (1)(0) + (2)(4) = -15 + 0 + 8 = -7$$

Q69. $A = \begin{bmatrix} 4 & 5 \\ -2 & -3 \end{bmatrix}$ then $A^{-1} = \underline{\hspace{2cm}}$

A. $\begin{bmatrix} -3 & -5 \\ 2 & -4 \end{bmatrix}$

B. $\frac{1}{2} \begin{bmatrix} -3 & -5 \\ 2 & 4 \end{bmatrix}$

C. $\frac{-1}{2} \begin{bmatrix} -4 & -5 \\ -2 & 3 \end{bmatrix}$

D. $\frac{-1}{2} \begin{bmatrix} -3 & -5 \\ 2 & 4 \end{bmatrix}$

Correct Ans: D

Solution :

$$A = \begin{bmatrix} 4 & 5 \\ -2 & -3 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad \text{for } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\therefore \det(A) = ad - bc = (4)(-3) - (5)(-2) = -12 + 10 = -2$$

$$\therefore A^{-1} = \frac{1}{-2} \begin{bmatrix} -3 & -5 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & \frac{5}{2} \\ -1 & -2 \end{bmatrix}$$

Q70. $225^\circ = \underline{\hspace{1cm}}$ *radian*.

- A. $\frac{3\pi}{4}$
- B. $\frac{5\pi}{4}$
- C. $\frac{7\pi}{4}$
- D. $\frac{9\pi}{4}$

Correct Ans: B

Solution :

$$\text{Radians} = \text{Degrees} \times \frac{\pi}{180}$$

$$\Rightarrow 225^\circ = 225 \times \frac{\pi}{180} = \frac{225\pi}{180} = \frac{5\pi}{4}$$

Q71. If $\sin \theta = \frac{2}{\sqrt{5}}$, then $\cot \theta = \underline{\hspace{1cm}}$

- A. 1
- B. $\frac{1}{2}$
- C. 2
- D. $\frac{1}{\sqrt{5}}$

Correct Ans: B

Solution :

$$\sin \theta = \frac{2}{\sqrt{5}}$$

$$\text{where, } \sin^2 \theta + \cos^2 \theta = 1$$

$$\therefore \left(\frac{2}{\sqrt{5}} \right)^2 + \cos^2 \theta = 1$$

$$\Rightarrow \frac{4}{5} + \cos^2 \theta = 1 \Rightarrow \cos^2 \theta = 1 - \frac{4}{5} = \frac{1}{5} \Rightarrow \cos \theta = \pm \frac{1}{\sqrt{5}}$$

$$\therefore \cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{\pm \frac{1}{\sqrt{5}}}{\frac{2}{\sqrt{5}}} = \pm \frac{1}{2}$$

Q72. Peripheral period of $\cos(3 + 4x) =$ _____

- A. $\frac{\pi}{4}$
- B. $\frac{-\pi}{4}$
- C. $\frac{\pi}{2}$
- D. $\frac{-\pi}{2}$

Correct Ans: C

Solution :

$$\text{Given : } \cos(3 + 4x)$$

$$\text{Period} = \frac{2\pi}{|B|} \quad \text{for } \cos(Bx + C)$$

→ The function is in the form $\cos(Bx + C)$, with:

$$\rightarrow B = 4$$

$$\rightarrow C = 3$$

$$\text{Period} = \frac{2\pi}{|4|} = \frac{2\pi}{4} = \frac{\pi}{2}$$

Q73. $|\hat{i} - 2\hat{j} - 2\hat{k}| =$ _____

- A. 3
- B. -3
- C. ± 3
- D. 9

Correct Ans: A

Solution :

$$\vec{A} = \hat{i} - 2\hat{j} - 2\hat{k}$$

$$|\vec{A}| = \sqrt{a^2 + b^2 + c^2}$$

$$|\vec{A}| = \sqrt{1^2 + (-2)^2 + (-2)^2} = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$$

Q74. $(\hat{i} \times \hat{j}) \circ \hat{k} = \underline{\hspace{2cm}}$

- A. $\hat{i}$
- B. $\hat{k}$
- C. 0
- D. 1

Correct Ans: D

Solution :

Given : $(\hat{i} \times \hat{j}) \circ \hat{k}$

Cross product:

$$\hat{i} \times \hat{j} = \hat{k}$$

Dot product:

$$\hat{k} \cdot \hat{k} = 1$$

$$(\hat{i} \times \hat{j}) \circ \hat{k} = 1$$

Q75. If the lines $x - 2y = 3$ and $3y + px - 1 = 0$ are perpendicular to each other then $p = \underline{\hspace{2cm}}$

- A. $\frac{3}{2}$
- B. $\frac{-3}{2}$
- C. 6
- D. -6

Correct Ans: C

Solution :

Given:

1. $x - 2y = 3$

2. $3y + px - 1 = 0$

$$x - 2y = 3 \Rightarrow -2y = -x + 3 \Rightarrow y = \frac{1}{2}x - \frac{3}{2}$$

$\therefore$ slope $m_1 = \frac{1}{2}$

Line 2:

$$3y + px = 1 \Rightarrow 3y = -px + 1 \Rightarrow y = \frac{-p}{3}x + \frac{1}{3}$$

$\therefore$ slope $m_2 = \frac{-p}{3}$

$\Rightarrow$ The condition for perpendicular lines:

$$m_1 \cdot m_2 = -1$$

$$\frac{1}{2} \cdot \left(\frac{-p}{3} \right) = -1 \Rightarrow \frac{-p}{6} = -1 \Rightarrow p = 6$$

Q76. A letter of adjustment is generally written in response to _____.

- A. letter of inquiry
- B. request letter
- C. order letter
- D. complaint letter

Correct Ans: D

Solution :

⇒ A letter of adjustment is written in response to a **complaint letter** to address grievances or resolve issues.

Q77. Which of the following statement is correct about letter of inquiry?

- A. written to offer a product.
- B. written to seek details about product or service.
- C. written to start investigation..
- D. written to raise a complaint.

Correct Ans: B

Solution :

⇒ A letter of inquiry (B) seeks details about products/services, distinct from complaints or offers.

Q78. Which of the following is an essential part of business letter?

- A. Executive summary
- B. Identification initials.
- C. Inside address
- D. Carbon copy notation

Correct Ans: C

Solution :

⇒ The **inside address** (recipient's address) is essential in business letters for formal communication.

Q79. Which of the following parts is used to take a courteous leave from a letter?

- A. Greetings
- B. Salutation
- C. Complimentary close
- D. Enclosure

Correct Ans: C

Solution :

⇒ The **complimentary close** (e.g., "Sincerely") is the courteous farewell in a letter.

Q80. Give the full form of 'PS' used in letter writing.

- A. Please
- B. Post Service
- C. Personnel Service
- D. Post Script

Correct Ans: D

Solution :

⇒ "PS" stands for **Post Script**, used to add information after the main message.

Q81. The talented singer danced quickly. (Identify the adverb)

- A. talented
- B. quickly
- C. danced
- D. singer

Correct Ans: B

Solution :

⇒ The adverb **quickly** modifies the verb "danced," describing how the action was performed.

Q82. The captain along with ten players _____ arrived at the stadium.
(Select the correct option)

- A. has
- B. have
- C. is
- D. are

Correct Ans: A

Solution :

⇒ With "captain along with," the singular verb **has** agrees with the subject "captain."

Q83. None of the products _____ up to the mark, so we returned them all.
(Select the correct option)

- A. is
- B. are
- C. was
- D. were

Correct Ans: D

Solution :

⇒ "None" with plural "products" requires plural verb **were** in past tense.

Q84. There is a church _____ the lake. (Select the correct option)

- A. over
- B. on
- C. between
- D. above

Correct Ans: B

Solution :

⇒ The church is located **on** the lake's edge (surface contact), not above/over.

Q85. He _____ to the gym every morning before work. (Select the correct option)

- A. goes
- B. went
- C. is going
- D. go

Correct Ans: A

Solution :

⇒ Present simple tense **goes** indicates habitual action ("every morning").

Q86. Select the correct spelling.

- A. Fahrenhit
- B. Ferrenhit
- C. Fahrenheit
- D. Fahrenheat

Correct Ans: C

Solution :

⇒ Correct spelling: **Fahrenheit**, named after physicist Daniel Fahrenheit.

Q87. Select the correct spelling.

- A. Inconvenience
- B. Inconveninse
- C. Inconvinience
- D. Inconviniance

Correct Ans: A

Solution :

⇒ Correct spelling: **Inconvenience** (meaning "trouble" or "hardship").

Q88. Identify the correct sentence.

- A. The criminals arrested by police.
- B. The police are arrested the criminals.
- C. The police were arrested the criminals.
- D. The police arrested the criminals.

Correct Ans: D

Solution :

⇒ Active voice correct: **"The police arrested the criminals"**. Others are passive/grammatically incorrect.

Q89. Identify the correct sentence.

- A. Shreya is tired as she had worked since 7 o'clock.
- B. Shreya is tired as she has been working since 7 o'clock.
- C. Shreya was tired as she has worked since 7 o'clock.
- D. Shreya is tired as she working since 7 o'clock.

Correct Ans: B

Solution :

⇒ Present perfect continuous **"has been working"** shows ongoing action from past to present.

Q90. Identify the correct sentence.

- A. The father purchased a vividly illustrated children's dictionary for his daughter.
- B. The father purchased a child's vividly illustrated dictionary for her daughter.
- C. The father purchased a children's vividly illustrated dictionary for his daughter.
- D. The father purchased an illustrated child's vivid dictionary for his daughter.

Correct Ans: A

Solution :

⇒ Correct phrasing: "vividly illustrated children's dictionary" (proper adjective-noun order).

Q91. Read the following passage carefully and answer the questions:

The ancient stepwell, Chand Baori, descended into the earth like a forgotten dream. Its geometric steps, thousand precise cuts into the Sandstone, spiraled down towards the emerald depths of the water. The air within its shadowed walls was cool, a welcome respite from the scorching Rajasthani sun. Villagers, their colourful turbans and saris vibrant against the stone, gathered at the upper levels, drawing water in earthen pots. Legends whispered of a hidden chamber deep within the stepwell, a place where the spirits of the desert dwelled, guarding secrets of forgotten Kings.

A traveling musician, his sitar echoing through the vast space, played haunting ragas, his melodies blending with the gentle splash of water. A lone scholar, his spectacles perched on his nose, meticulously studied the intricate carvings on the walls, searching for clues to the stepwell's mysterious origins and the stories etched into its sandstone heart.

What was the purpose of the steps in Chand Baori?

- A. To provide seating for spectators
- B. To create a maze for tourists
- C. To allow access to the water deep below
- D. To form a decorative pattern

Correct Ans: C

Solution :

⇒ Steps allow access to water in the stepwell, as described in the passage.

Q92. The "cool shadowed walls" of the stepwell provide:

- A. A place for loud celebrations.
- B. A relief from the desert heat.
- C. A place for modern art display.
- D. A site for industrial activity.

Correct Ans: B

Solution :

⇒ "Cool shadowed walls" provide relief from desert heat, contrasting the "scorching sun."

Q93. According to legends, what was hidden deep within the stepwell?

- A. A treasure vault
- B. Sacred water
- C. A secret passage to the fort
- D. A chamber of desert spirit

Correct Ans: D

Solution :

⇒ Legends mention a **chamber of desert spirits** guarding secrets.

Q94. The phrase "forgotten dream" used to describe Chand Baori suggests:

- A. The stepwell's historical significance and potential obscurity.
- B. The stepwell's modern architectural design.
- C. The stepwell's role as a tourist attraction.
- D. The stepwell's frequent use by villagers.

Correct Ans: A

Solution :

⇒ "Forgotten dream" implies **historical significance and obscurity**, not modern use.

Q95. What was the scholar searching for in the carvings?

- A. Precious gems
- B. Ancient map of desert
- C. Clues to stepwell's origin
- D. Hidden doorways

Correct Ans: C

Solution :

⇒ The scholar studies carvings for **clues to the stepwell's origin**.

Q96. While communicating, the process of understanding the original idea of the message is called _____.

- A. Decoding
- B. Feedback
- C. Encoding
- D. Transmission

Correct Ans: A

Solution :

⇒ **Decoding** is interpreting the sender's message by the receiver.

Q97. _____ is the study of body language and its effect on communication including facial expression, gestures, etc.

- A. Proxemics
- B. Chronemics
- C. Paralanguage
- D. Kinesics

Correct Ans: D

Solution :

⇒ **Kinesics** studies body language (gestures, expressions).

Q98. Which of the following is not an example of verbal communication ?

- A. A manager giving instructions to workers orally.
- B. A man shaking hand with a foreigner.
- C. A student writing a letter to the principal.
- D. An oldman reading a news paper.

Correct Ans: B

Solution :

⇒ **Shaking hands** is non-verbal; others involve spoken/written words.

Q99. What is common result of socio-cultural barrier to communication?

- A. Enhanced comprehension.
- B. Improved content clarity.
- C. Misinterpretation of intent.
- D. Enriched social connection.

Correct Ans: C

Solution :

⇒ Socio-cultural barriers often cause **misinterpretation** due to differing norms.

Q100. Factors such as person's emotions, anxiety, stress and fear can form _____ barrier to communication.

- A. Personal Barrier
- B. Psychological Barrier
- C. Physical Barrier
- D. Neurological Barrier

Correct Ans: B

Solution :

⇒ Emotions like anxiety create **psychological barriers** to effective communication.

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